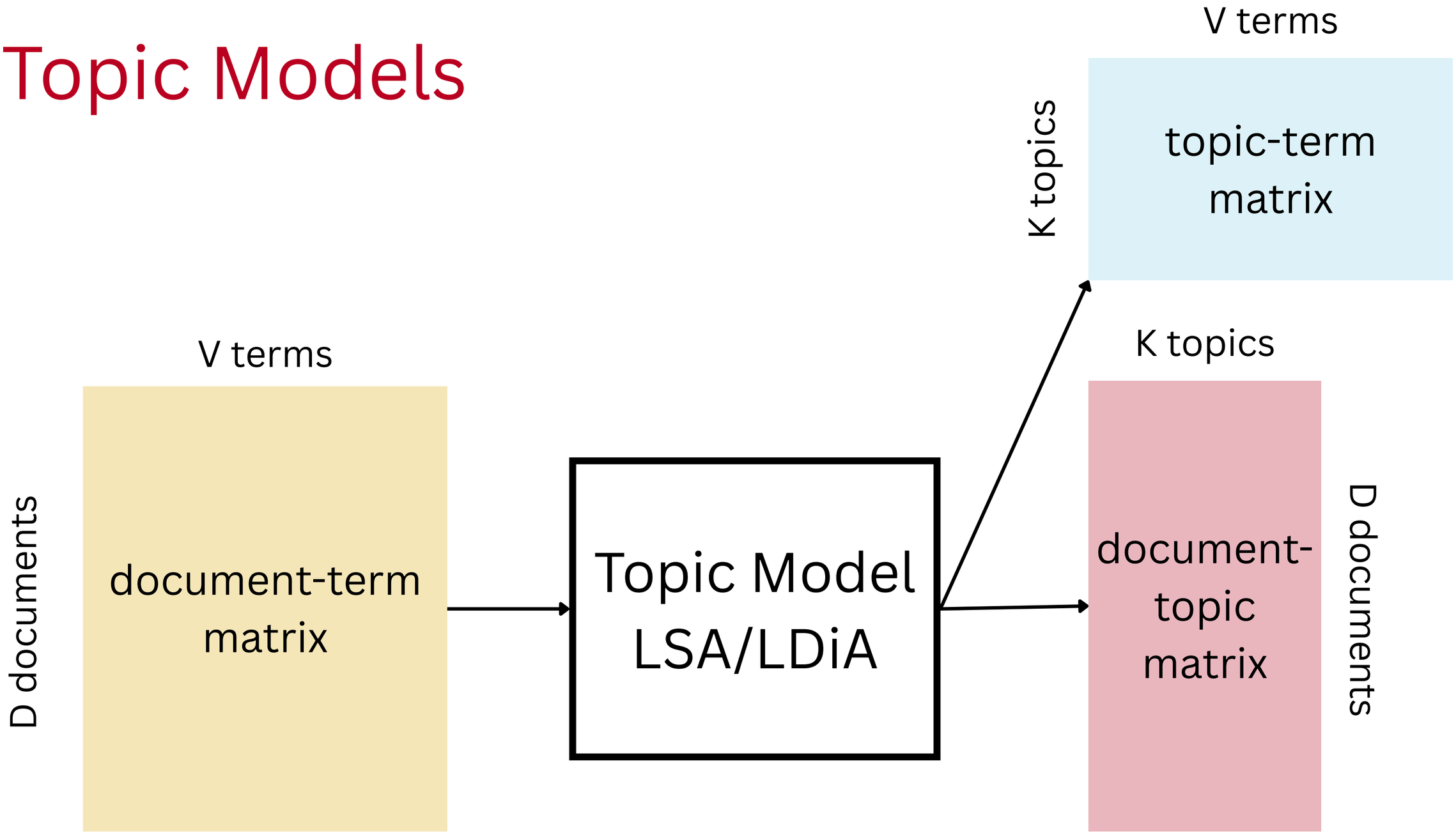
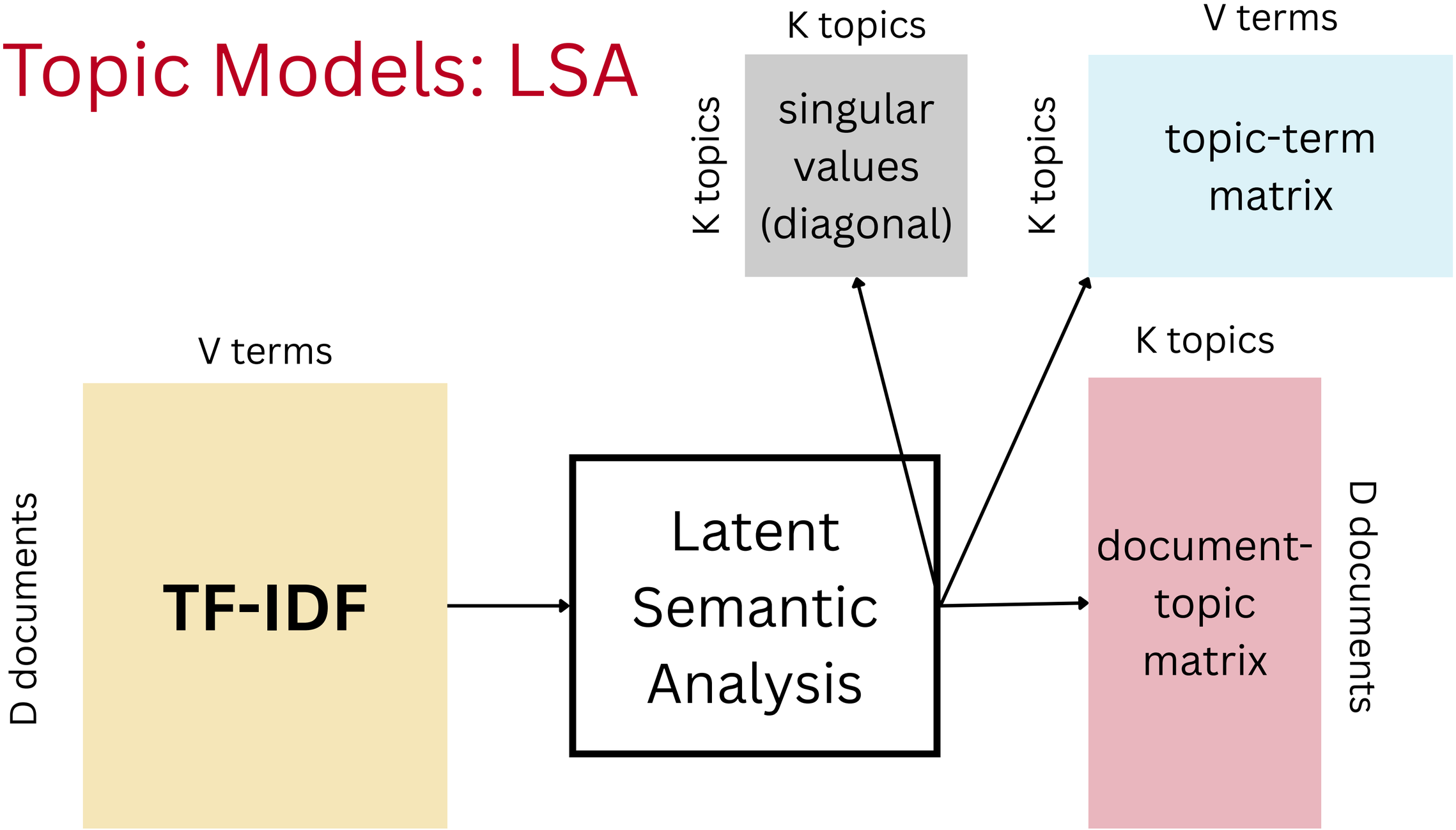
Natural Language Processing
Lab 6: Topic Models LSA, LDiA

Lonny Chen: l.chen@hertie-school.org
GRAD-E1282, Fall 2026
7-8 October 2026

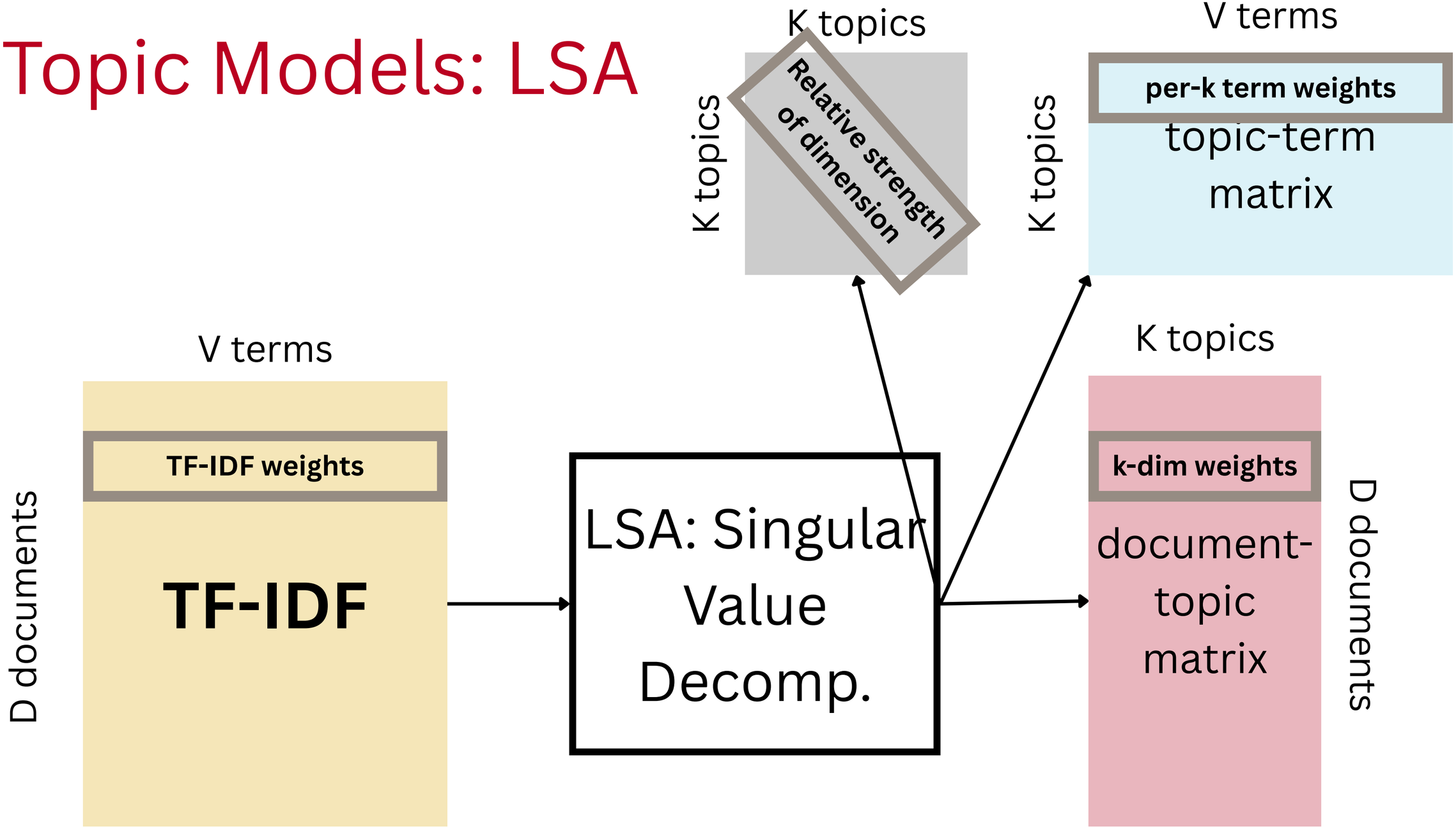
Topic Models



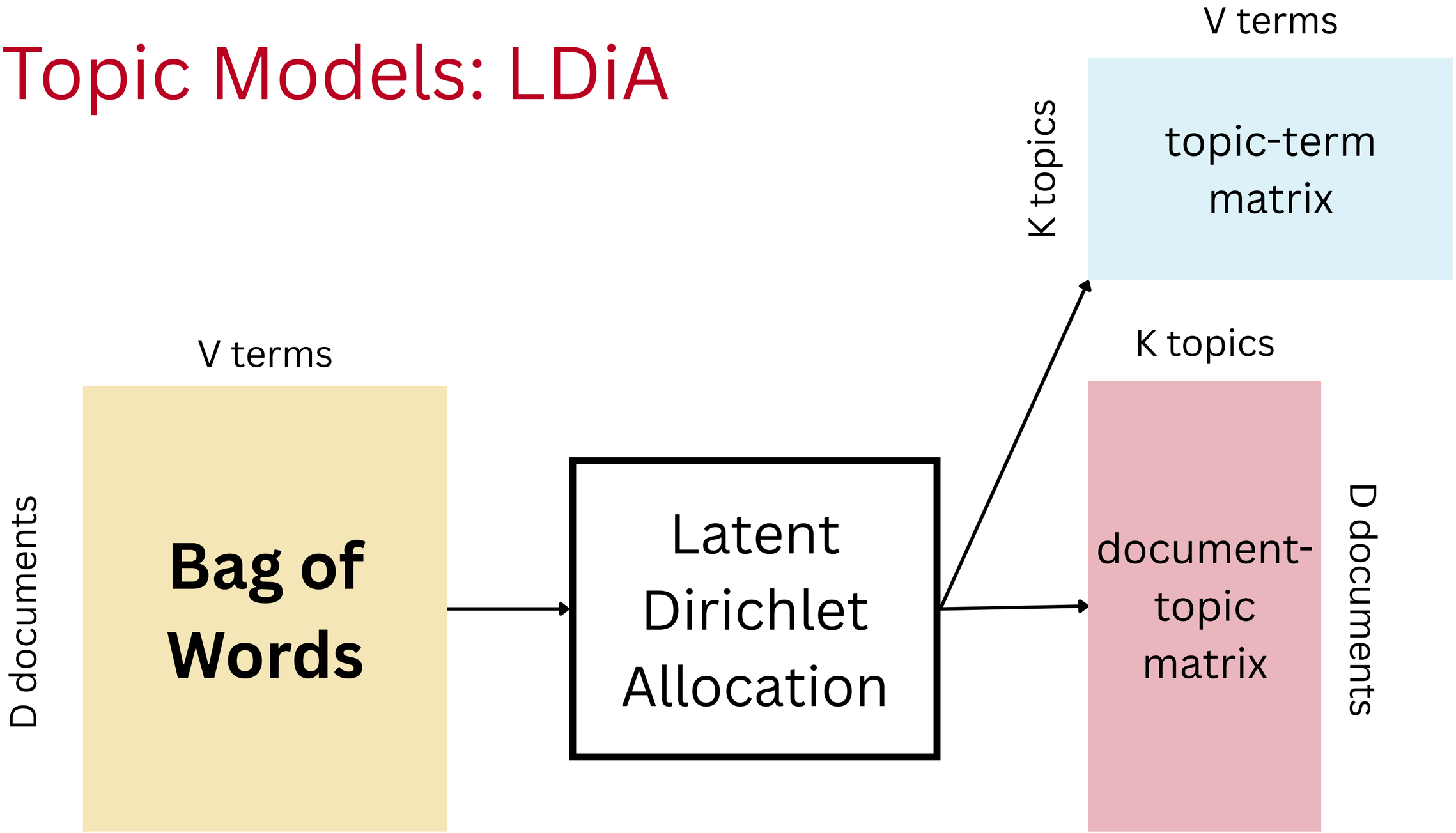
Topic Models: LSA



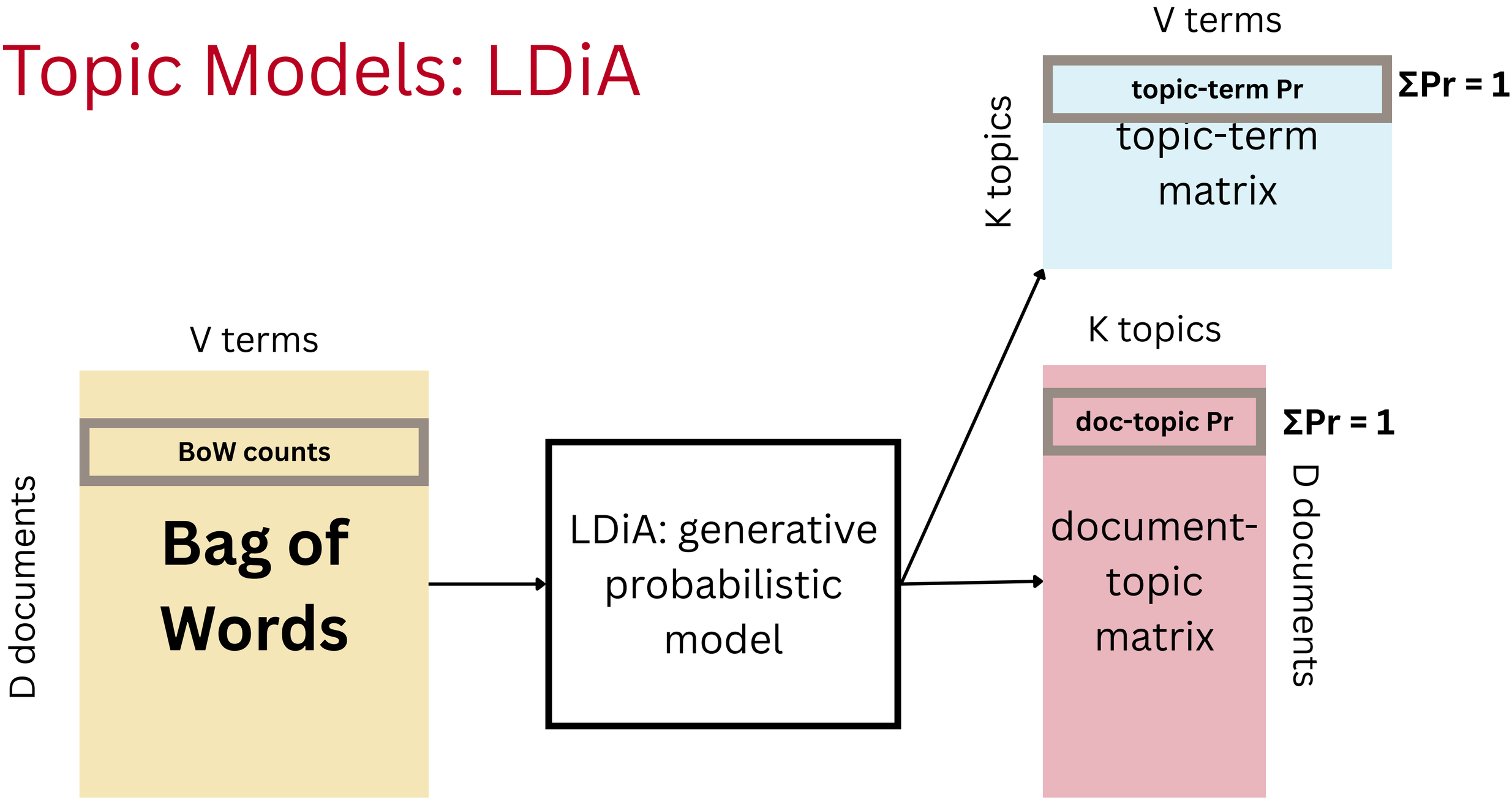
Topic Models: LSA



Topic Models: LDiA



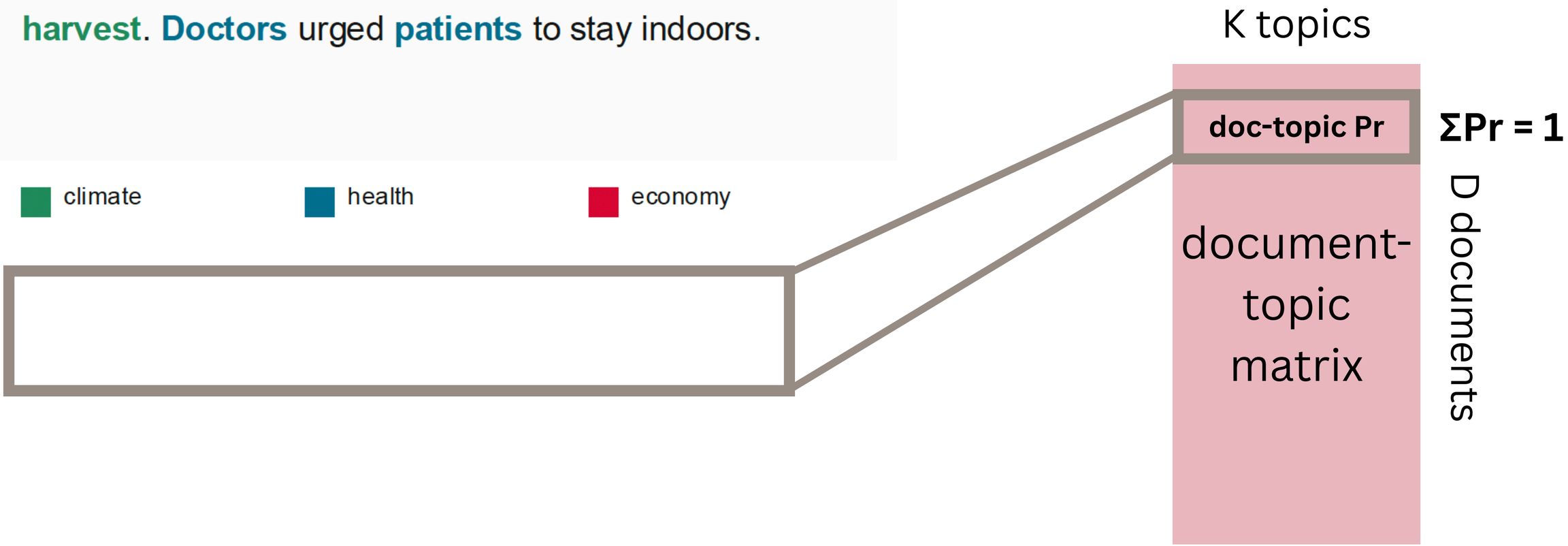
Topic Models: LDiA



LDiA: Document-Topic Distribution

From Session 6:

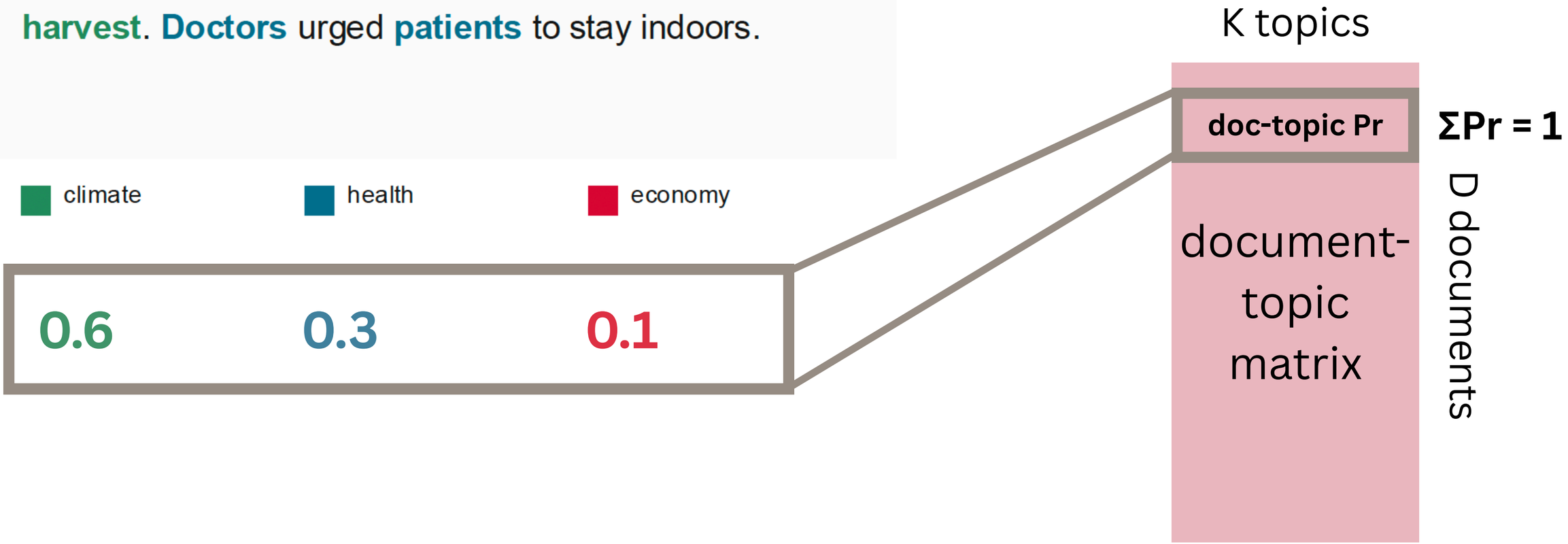
The **heatwave** pushed **hospital admissions** up as **energy prices** hit records and **drought** hurt the **harvest**. **Doctors** urged **patients** to stay indoors.



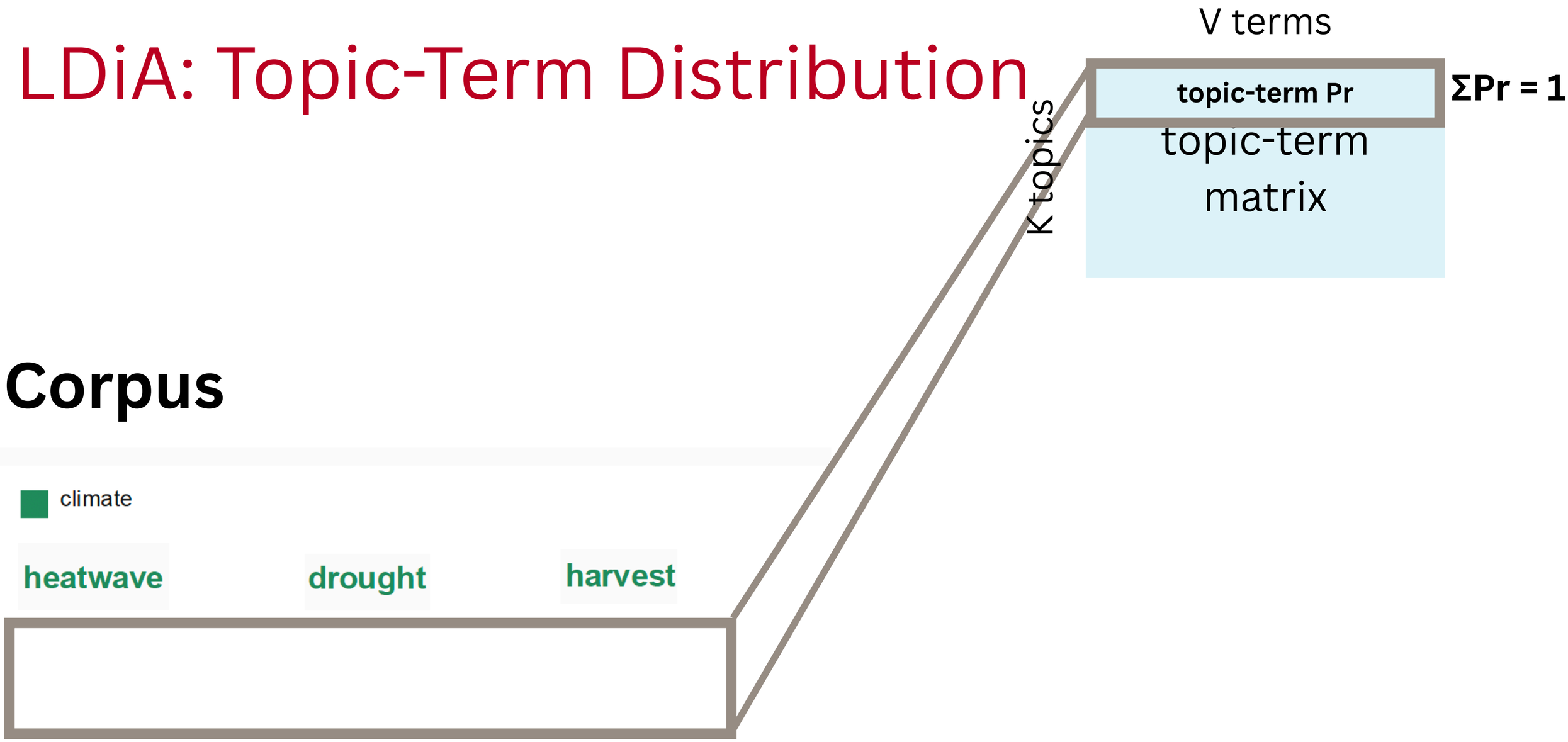
LDiA: Document-Topic Distribution

From Session 6:

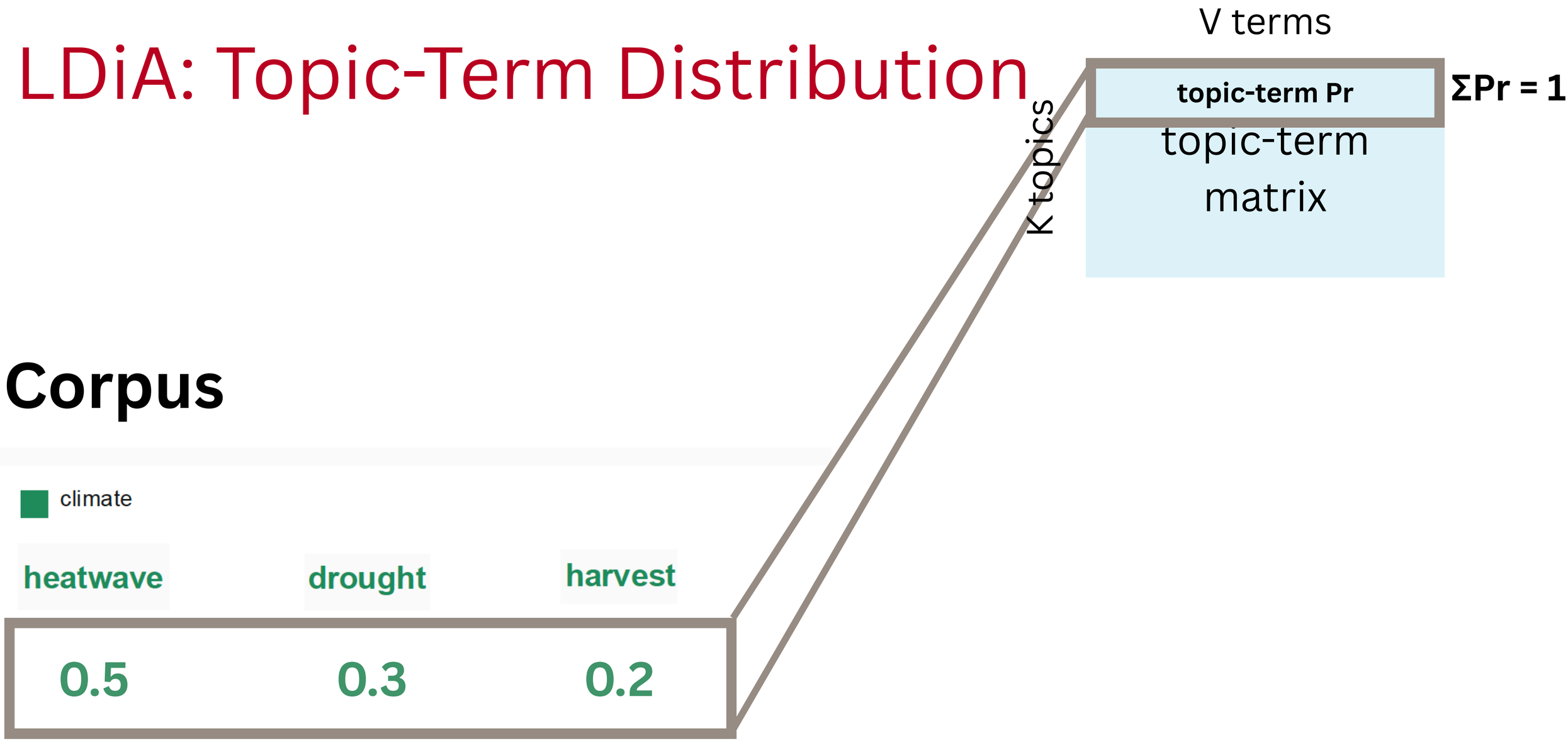
The **heatwave** pushed **hospital admissions** up as **energy prices** hit records and **drought** hurt the **harvest**. **Doctors** urged **patients** to stay indoors.



LDiA: Topic-Term Distribution



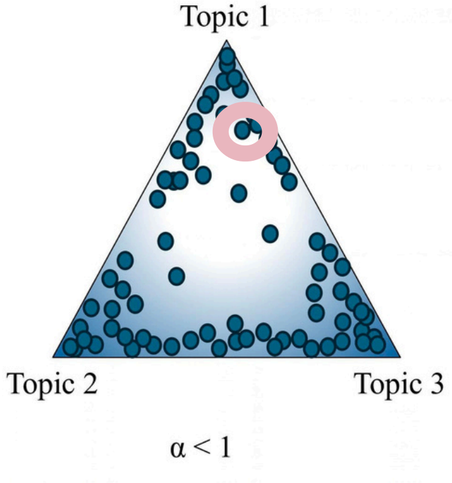
LDiA: Topic-Term Distribution

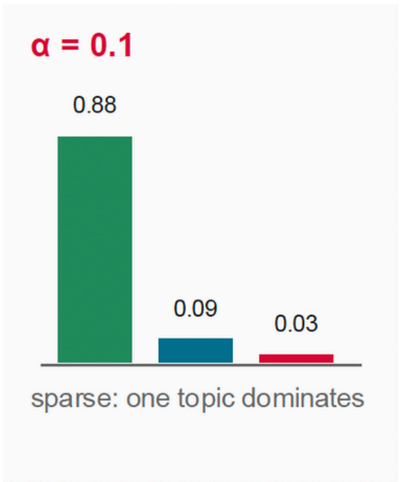


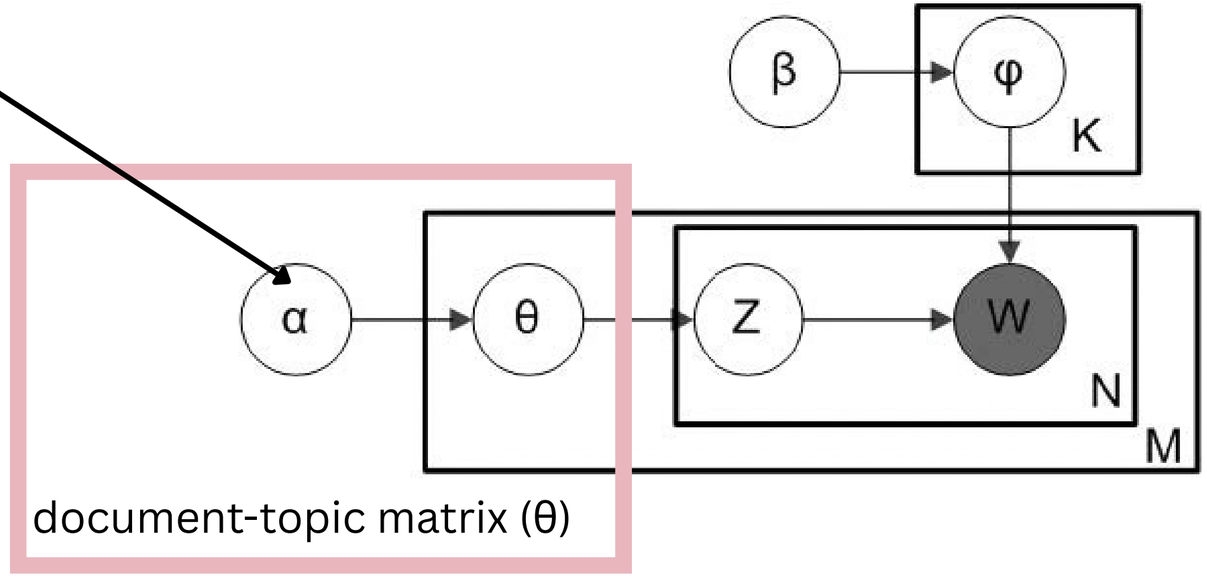
LDiA: Document “Generation” Process

~ Dirichlet(prior)

- Prior parameter “α”
  - Low < 1: sparse Pr, favour dominant topic
- Output: list of topic probabilities per-document





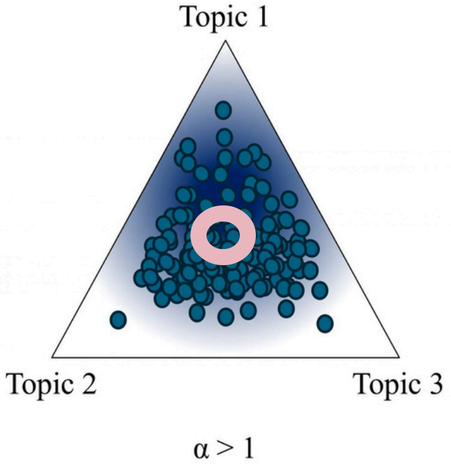


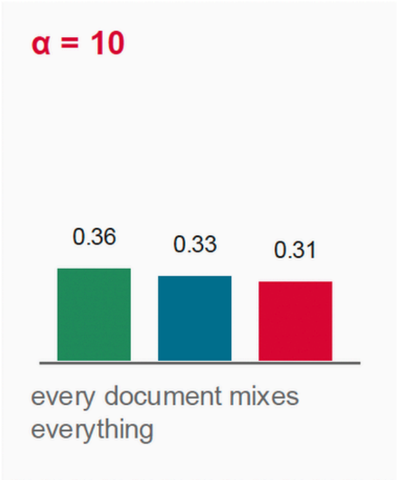
Source: *Pérez & Solano “A Comprehensive Guide to Latent Dirichlet Allocation“ (2025)*

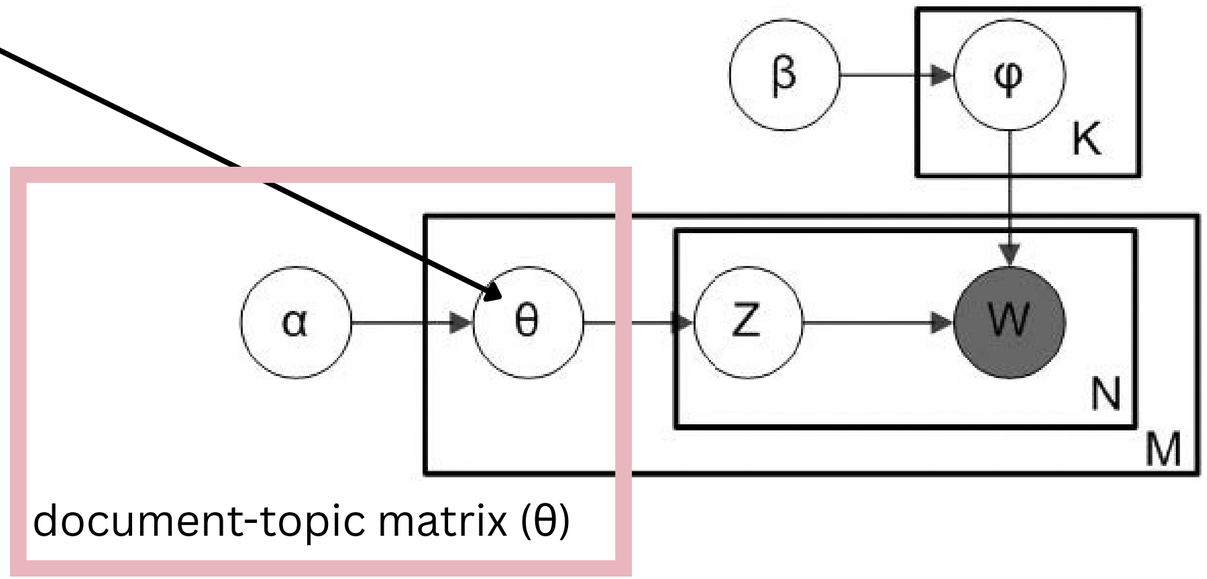
LDiA: Document “Generation” Process

~ Dirichlet(prior)

- Prior parameter “α”
  - High > 1: dense Pr, favour mixed topics
- Output: list of topic probabilities per-document





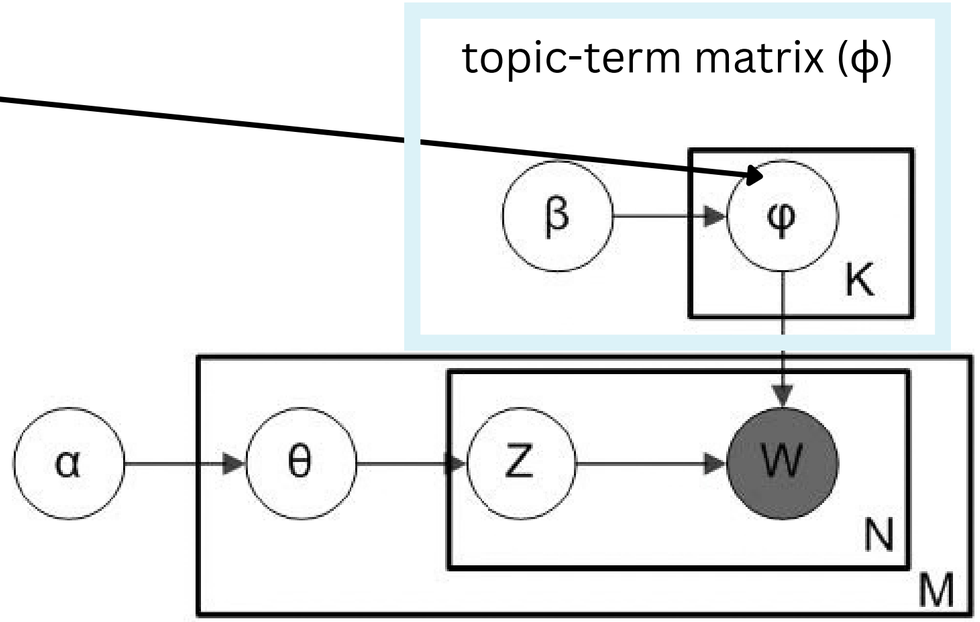


Source: *Pérez & Solano “A Comprehensive Guide to Latent Dirichlet Allocation“ (2025)*

LDiA: Document “Generation” Process

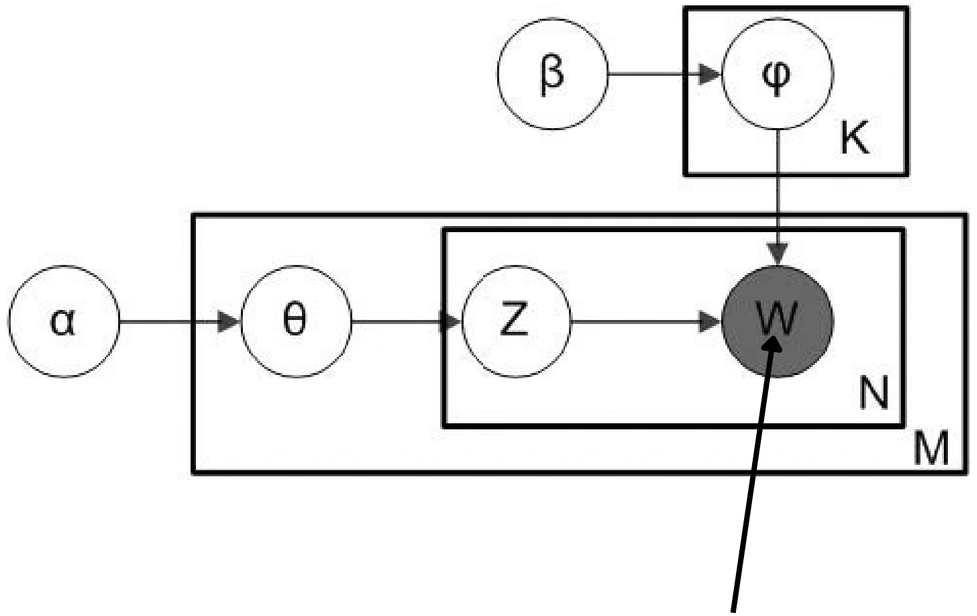
~ Dirichlet(prior)

- Prior parameter “β”
- Output: list of term probabilities per-topic



Source: *Pérez & Solano “A Comprehensive Guide to Latent Dirichlet Allocation“ (2025)*

LDiA: Document “Generation” Process

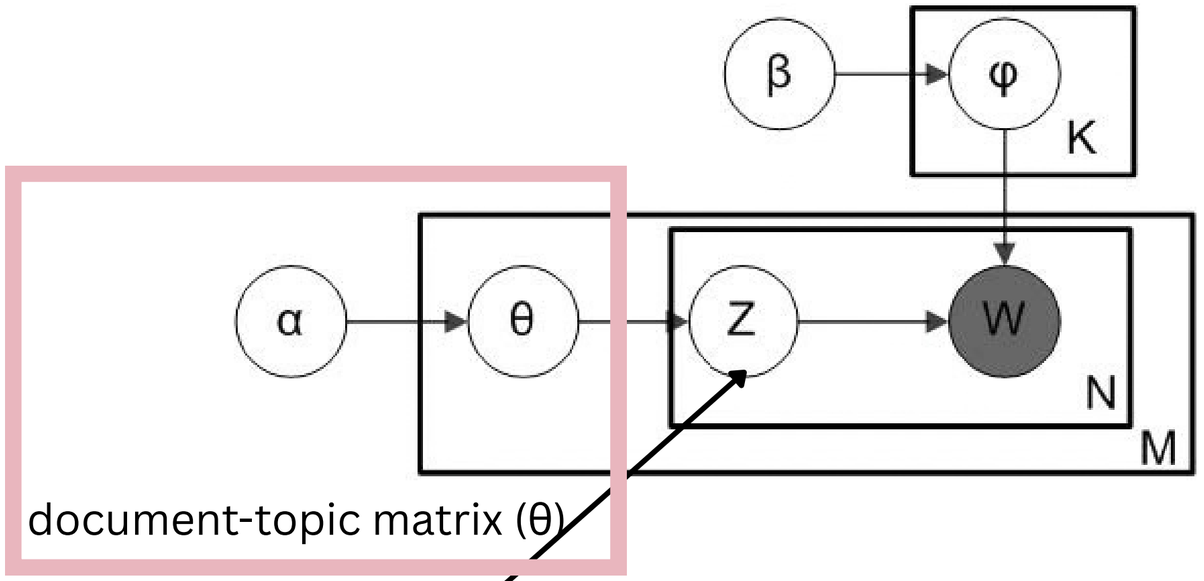


Each document-word n

- Assign a topic to each word

Source: *Pérez & Solano “A Comprehensive Guide to Latent Dirichlet Allocation“ (2025)*

LDiA: Document “Generation” Process



~ Multinomial (θ)

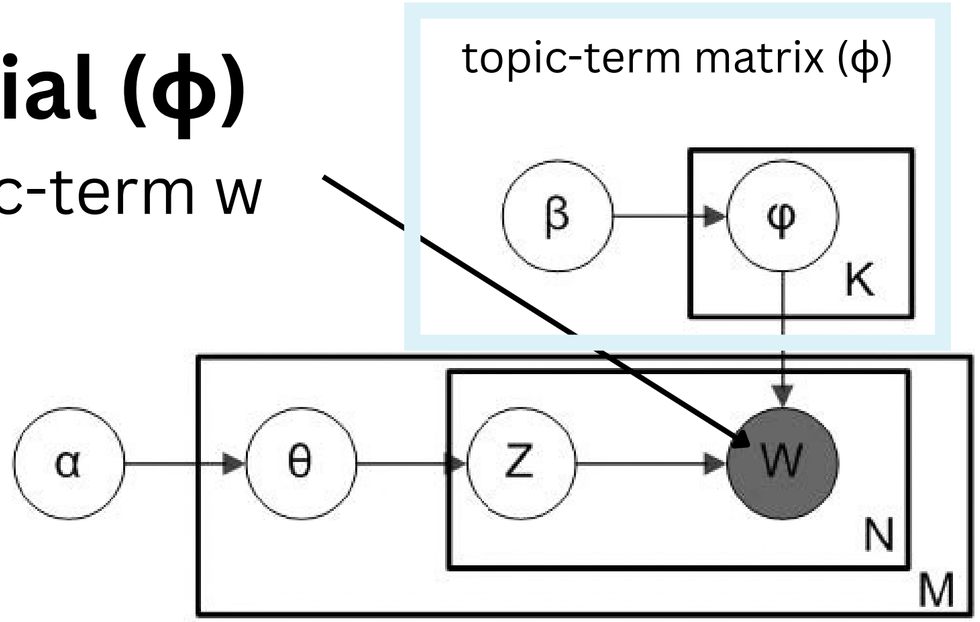
- Draw a document-topic z for this word from θ

Source: *Pérez & Solano “A Comprehensive Guide to Latent Dirichlet Allocation“ (2025)*

LDiA: Document “Generation” Process

~ Multinomial (ϕ)

- Draw a topic-term w from ϕ



Source: *Pérez & Solano “A Comprehensive Guide to Latent Dirichlet Allocation“ (2025)*

Next Week: BERTopic, Neural Models

